

Description and choice data for the domain “Radio formats”

Description of the choice domain 10, Radio formats

The prompt question and the universe of five response options in the choice domain **Radio formats** are as follows. The labels *a*, *b*, *c*, *d* and *e* were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

The domain is radio formats, and the choice objects are the top 5 radio formats in Canada in 2015, according to

<https://byrnesmedia.com/2015/10/05/the-6-best-performing-radio-formats-in-canada/>

Descriptions of formats are from

<http://www.newsgeneration.com/broadcast-resources/guide-to-radio-station-formats/>

Suppose you were on a two hour road trip and you have a choice among radio stations with the following formats. Which one would you choose?

- News
- Hot Adult Contemporary, or Hot AC (A variety of classic and contemporary mainstream music geared towards adults.)
- Classic Hits (Rock and pop, roughly 1964-1989)
- Country Music
- Adult Contemporary, or AC (Adult-oriented pop/rock with no hard rock.)

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this domain.

The screenshot shows a web interface for a choice experiment. At the top, the title "Radio formats" is displayed in blue. Below it, the prompt "Suppose you were on a two hour road trip and you have a choice among radio stations with the following formats. Which one would you choose?" is shown. Two radio button options are listed: "Adult Contemporary, or AC (Adult-oriented pop/rock with no hard rock.)" and "Country Music". A progress bar is visible on the right side of the menu, and a blue button with the text ">>" is located at the bottom right.

Figure 1: Screenshot for domain Radio formats

Menu A	Choice counts					Choice proportions				
	$N_A(a)$	$N_A(b)$	$N_A(c)$	$N_A(d)$	$N_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	10	30	-	-	-	0.250	0.750	-	-	-
$\{a, c\}$	11	-	29	-	-	0.275	-	0.725	-	-
$\{b, c\}$	-	13	27	-	-	-	0.325	0.675	-	-
$\{a, b, c\}$	5	4	31	-	-	0.125	0.100	0.775	-	-
$\{a, d\}$	22	-	-	18	-	0.550	-	-	0.450	-
$\{b, d\}$	-	28	-	12	-	-	0.700	-	0.300	-
$\{a, b, d\}$	9	22	-	9	-	0.225	0.550	-	0.225	-
$\{c, d\}$	-	-	39	1	-	-	-	0.975	0.025	-
$\{a, c, d\}$	3	-	32	5	-	0.075	-	0.800	0.125	-
$\{b, c, d\}$	-	7	26	7	-	-	0.175	0.650	0.175	-
$\{a, b, c, d\}$	3	8	24	5	-	0.075	0.200	0.600	0.125	-
$\{a, e\}$	15	-	-	-	25	0.375	-	-	-	0.625
$\{b, e\}$	-	16	-	-	24	-	0.400	-	-	0.600
$\{a, b, e\}$	10	12	-	-	18	0.250	0.300	-	-	0.450
$\{c, e\}$	-	-	34	-	6	-	-	0.850	-	0.150
$\{a, c, e\}$	4	-	25	-	11	0.100	-	0.625	-	0.275
$\{b, c, e\}$	-	7	26	-	7	-	0.175	0.650	-	0.175
$\{a, b, c, e\}$	2	10	21	-	7	0.050	0.250	0.525	-	0.175
$\{d, e\}$	-	-	-	9	31	-	-	-	0.225	0.775
$\{a, d, e\}$	8	-	-	9	24	0.195	-	-	0.220	0.585
$\{b, d, e\}$	-	12	-	9	20	-	0.293	-	0.220	0.488
$\{a, b, d, e\}$	8	12	-	6	14	0.200	0.300	-	0.150	0.350
$\{c, d, e\}$	-	-	22	6	12	-	-	0.550	0.150	0.300
$\{a, c, d, e\}$	7	-	19	11	3	0.175	-	0.475	0.275	0.075
$\{b, c, d, e\}$	-	9	19	4	8	-	0.225	0.475	0.100	0.200
$\{a, b, c, d, e\}$	3	5	22	2	8	0.075	0.125	0.550	0.050	0.200

Table 1: Observed choice counts and proportions.

Choice data for domain 10, Radio formats

Table 1 shows choice counts and choice proportions for this choice domain. For each menu A and each object $x \in \{a, b, c, d, e\}$, $N_A(x)$ is the number of participants who chose object x from menu A and $\hat{P}_A(x)$ is the corresponding proportion of participants who chose x from A . When $x \notin A$, a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of $\{a, b, c, d, e\}$. The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.

